

Relativistic Quantum Field Theory I(PH.60053)

Exercise Set 1

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Note: I generally disdain the use of natural units in undergraduate studies. For students having gone through a symbolic mess of a calculation, something as simple as dimensional analysis gives a good sanity check. And while this course is explicitly designed for graduate students, I would like to stand by this conviction and perform all calculations with explicit engineering dimensions and constants. I believe it is reasonably simple for the grader to convert it into natural units. I appreciate your understanding.

Problem 1

We use the following definitions:

$$A^\mu = \left(\frac{\phi}{c}, \vec{A}\right), \quad F^{\mu\nu} = \partial^{[\mu} A^{\nu]}, \quad J^\mu = (-c\rho, \vec{J}), \quad \partial_\mu = \left(\frac{1}{c}\partial_t, \vec{\nabla}\right), \quad \partial^\mu = \left(-\frac{1}{c}\partial_t, \vec{\nabla}\right),$$

$$E^i = -\partial_i \phi - \partial_t A^i = c \left(\left(-\frac{1}{c}\partial_t\right) A^i - \partial_i \left(\frac{\phi}{c}\right) \right) = cF^{0i},$$

$$B^i = \partial_j A^k - \partial_k A^j = F^{jk} \quad (\epsilon_{ijk} = +1).$$

We start with the Lagrangian density:

$$\mathcal{L} = -\frac{1}{4\mu_0} F_{\alpha\beta} F^{\alpha\beta} + J_\alpha A^\alpha.$$

We find the following partial derivative relations:

$$\frac{\partial(\partial_\alpha A_\beta)}{\partial(\partial_\mu A_\nu)} = \delta_\alpha^\mu \delta_\beta^\nu \Rightarrow \frac{\partial(\partial^\alpha A^\beta)}{\partial(\partial_\mu A_\nu)} = \eta^{\mu\alpha} \eta^{\nu\beta}$$

$$\Rightarrow \frac{\partial}{\partial(\partial_\mu A_\nu)} (F_{\alpha\beta} F^{\alpha\beta}) = (\delta_\alpha^\mu \delta_\beta^\nu - \delta_\beta^\mu \delta_\alpha^\nu) F^{\alpha\beta} + F_{\alpha\beta} (\eta^{\mu\alpha} \eta^{\nu\beta} - \eta^{\mu\beta} \eta^{\nu\alpha}) = 4F^{\mu\nu}$$

Thus, the Euler-Lagrange equations are given by

$$0 = \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu A_\nu)} \right) - \frac{\partial \mathcal{L}}{\partial A_\nu} = \partial_\mu \left(-\frac{1}{\mu_0} F^{\mu\nu} \right) - J^\nu$$

$$\therefore \partial_\mu F^{\mu\nu} = -\mu_0 J^\nu$$

$$1. \quad \nu = 0: \partial_\mu F^{\mu 0} = -\frac{1}{c} \vec{\nabla} \cdot \vec{E} \Rightarrow \vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}$$

$$2. \nu = 1: \partial_\mu F^{\mu 1} = \left(\frac{1}{c}\partial_t\right)\left(\frac{E_x}{c}\right) + \partial_y(-B^z) + \partial_z B^y = \left(\frac{1}{c^2}\partial_t \vec{E} - \vec{\nabla} \times \vec{B}\right)^x.$$

Thus, generally, $\vec{\nabla} \times \vec{B} = \mu_0 \vec{J} + \frac{1}{c^2}\partial_t \vec{E}$.

Furthermore, the Bianchi identity can be expressed as follows:

$$1. \partial_{[0}F_{12]} = \frac{1}{c}\partial_t B^z + \partial_x\left(+\frac{E_y}{c}\right) + \partial_y\left(-\frac{E_x}{c}\right) = \frac{1}{c}\left(\partial_t \vec{B} + \vec{\nabla} \times \vec{E}\right)^z.$$

Thus, generally, $\vec{\nabla} \times \vec{E} = -\partial_t \vec{B}$.

$$2. \partial_{[1}F_{23]} = \partial_x B^x + \partial_y B^y + \partial_z B^z = \vec{\nabla} \cdot \vec{B} = 0.$$

Thus, we see that the Euler-Lagrange equations as well as the Bianchi identity successfully reproduce the conventional Maxwell equations. $\square$

Problem 2

From the special-relativistic identity $E^2 = \sqrt{p^2 c^2 + m^2 c^4}$, we attempt to define the Hamiltonian operator as:

$$H \stackrel{?}{=} \sqrt{-\hbar^2 c^2 \vec{\nabla}^2 + m^2 c^4} = mc^2 \sqrt{1 - \left(\frac{\hbar}{mc} \vec{\nabla}\right)^2}.$$

To transform this into a kernel function, i.e., find the function $K(\vec{x}, \vec{y})$ such that

$$(H\psi)(\vec{x}) = \int d^3\vec{y} K(\vec{x}, \vec{y}) \psi(\vec{y}),$$

we utilize the Fourier transform:

$$\psi(\vec{x}) = \int d^3\vec{k} \psi_{\vec{k}} e^{i\vec{k} \cdot \vec{x}}, \quad \psi_{\vec{k}} = \int \frac{d^3\vec{x}}{(2\pi)^3} \psi(\vec{x}) e^{-i\vec{k} \cdot \vec{x}}.$$

Substitution then yields:

$$\begin{aligned} (H\psi)(\vec{x}) &= mc^2 \int d^3\vec{k} \psi_{\vec{k}} \sqrt{1 - \left(\frac{\hbar}{mc} \vec{\nabla}\right)^2} e^{i\vec{k} \cdot \vec{x}} \\ &= mc^2 \int d^3\vec{k} \psi_{\vec{k}} \sqrt{1 - \left(\frac{\hbar\vec{k}}{mc}\right)^2} e^{i\vec{k} \cdot \vec{x}} \\ &= mc^2 \int d^3\vec{k} \left(\int \frac{d^3\vec{y}}{(2\pi)^3} \psi(\vec{y}) e^{-i\vec{k} \cdot \vec{y}} \right) \sqrt{1 - \left(\frac{\hbar\vec{k}}{mc}\right)^2} e^{i\vec{k} \cdot \vec{x}} \\ &= \int d^3\vec{y} \left(mc^2 \int \frac{d^3\vec{k}}{(2\pi)^3} \sqrt{1 - \left(\frac{\hbar\vec{k}}{mc}\right)^2} e^{i\vec{k} \cdot (\vec{x} - \vec{y})} \right) \psi(\vec{y}) \end{aligned}$$

Thus, defining $\vec{r} := \vec{x} - \vec{y}$, we find that

$$\begin{aligned}
K(\vec{x}, \vec{y}) &= mc^2 \int \frac{d^3 \vec{k}}{(2\pi)^3} \sqrt{1 - \left(\frac{\hbar \vec{k}}{mc} \right)^2} e^{i \vec{k} \cdot \vec{r}} \\
&= \frac{m^4 c^5}{8\pi^3 \hbar^3} \int d^3 \vec{\kappa} \sqrt{1 + \vec{\kappa}^2} e^{i \vec{\kappa} \cdot \vec{\rho}} \quad \left(\vec{\kappa} := \frac{\hbar \vec{k}}{mc}, \quad \vec{\rho} := \frac{mc \vec{r}}{\hbar} \right) \\
&= \frac{m^4 c^5}{8\pi^3 \hbar^3} \int_0^\infty d\kappa \kappa^2 \sqrt{1 + \kappa^2} \int_{-1}^1 d(\cos \theta) e^{i \kappa \rho \cos \theta} \int_0^{2\pi} d\phi \\
&= \frac{m^4 c^5}{2\pi^2 \hbar^3 \rho} \int_0^\infty d\kappa \kappa \sqrt{1 + \kappa^2} \sin \kappa \rho.
\end{aligned}$$

Despite the obvious problem of convergence, we nevertheless attempt to sidestep the issue using appropriate Feynmann tricks.

$$\begin{aligned}
K(\vec{x}, \vec{y}) &= \frac{m^4 c^5}{2\pi^2 \hbar^3 \rho} \int_0^\infty d\kappa \left(\frac{\kappa^3}{\sqrt{1 + \kappa^2}} + \frac{\kappa}{\sqrt{1 + \kappa^2}} \right) \sin \kappa \rho \\
&= \frac{m^4 c^5}{2\pi^2 \hbar^3 \rho} \left(\frac{d^3}{d\rho^3} - \frac{d}{d\rho} \right) \int_0^\infty d\kappa \frac{\cos \kappa \rho}{\sqrt{1 + \kappa^2}} \\
&= \frac{m^4 c^5}{2\pi^2 \hbar^3 \rho} (K_0'''(\rho) - K_0'(\rho)),
\end{aligned}$$

where $K_0(x)$ is the modified Bessel function of the second kind of order 0. The last expression can be simplified using well-known recurrence relations:

$$\begin{aligned}
K_0'(x) &= -K_1(x), \quad K_1'(x) = -K_0(x) - \frac{1}{x} K_1(x), \quad K_2(x) = K_0(x) + \frac{2}{x} K_1(x). \\
&\Rightarrow K_0'''(x) - K_0'(x) = -\frac{1}{x} K_2(x) \\
\therefore K(\vec{x}, \vec{y}) &= -\frac{m^4 c^5}{2\pi^2 \hbar^3} \frac{K_2(\rho)}{\rho^2} = -\frac{m^2 c^3}{2\pi^2 \hbar} \frac{K_2\left(\frac{mc}{\hbar} r\right)}{r^2}
\end{aligned}$$

Notice the blatant nonlocality: $K(\vec{x}, \vec{y})$ has support arbitrarily far away from $\vec{x} = \vec{y}$! In fact, using the asymptotic behavior $K_\alpha(x) \stackrel{x \rightarrow \infty}{\sim} \sqrt{\frac{\pi}{2x}} e^{-x}$, we find that

$$K(\vec{x}, \vec{y}) \stackrel{r \rightarrow \infty}{\sim} -\sqrt{\frac{m^3 c^5}{8\pi^3 \hbar}} r^{-2} e^{-\frac{mc}{\hbar} r}.$$

Problem 3

We first need to reconstruct the omitted natural constants in the formulae. With the ground truths

$$[\mathcal{L}] = [EL^{-3}] \quad \text{and} \quad [\partial] = [L^{-1}],$$

we insist that

$$[\mathcal{L}] = [\partial^\mu \phi \partial_\mu \phi] = [\phi^2 \cdot L^{-2}] \Rightarrow [\phi] = [E^{1/2} L^{-1/2}].$$

(Note: I could not decide whether this is the most useful convention. I also think it would be reasonable to insist that ϕ is dimensionless and multiply by appropriate natural constants to match

the dimensions. This is just the convention that Gemini suggested, so it's its fault.) Moreover, we insist that $[m] = [M] = [EL^{-2}T^2]$, which implies

$$[m^2\phi^2] = [(EL^{-2}T^2)^2 \cdot EL^{-1}] = [E^3L^{-5}T^4] = [EL^{-3} \cdot (ET)^2 \cdot (LT^{-1})^{-2}] = [\mathcal{L} \cdot \hbar^2 c^{-2}].$$

Thus, the reconstructed Lagrangian density is given by

$$\mathcal{L} = \frac{1}{2} \left(-\partial_\mu \phi \partial^\mu \phi - \left(\frac{mc}{\hbar} \right)^2 \phi^2 \right).$$

Now, we can actually start solving the problem. The kinetic term can be expanded as:

$$\mathcal{L} = \frac{1}{2} \left(\left(\frac{\dot{\phi}}{c} \right)^2 - (\vec{\nabla} \phi)^2 - \left(\frac{mc}{\hbar} \right)^2 \phi^2 \right),$$

which lets us find the conjugate momentum field:

$$\pi(t, \vec{x}) := \frac{\partial \mathcal{L}}{\partial \dot{\phi}(t, \vec{x})} = \frac{\dot{\phi}}{c^2}.$$

Thus, the Hamiltonian functional is given by:

$$\begin{aligned} H[\phi, \pi; t] &= \int d^3 \vec{x} \left(\pi(t, \vec{x}) \dot{\phi}(t, \vec{x}) - \mathcal{L}[\phi, \dot{\phi}] \right) \\ &= \int d^3 \vec{x} \frac{1}{2} \left((c\pi(t, \vec{x}))^2 + (\vec{\nabla} \phi(t, \vec{x}))^2 + \left(\frac{mc}{\hbar} \phi(t, \vec{x}) \right)^2 \right). \end{aligned}$$

We now write down the Heisenber equations of motion:

$$\frac{dA}{dt} = -\frac{1}{i\hbar} [H[\phi, \pi; t], A].$$

The equation for $\phi(t, \vec{x})$ is relatively straightforward:

- $[\pi(t, \vec{y})^2, \phi(t, \vec{x})] = \{\pi(t, \vec{y}), [\pi(t, \vec{y}), \phi(t, \vec{x})]\} = -2i\hbar\delta^3(\vec{y} - \vec{x})\pi(t, \vec{y})$
- $[\vec{\nabla}_y^2 \phi(t, \vec{y}), \phi(t, \vec{x})] = \vec{\nabla}_y^2 [\phi(t, \vec{y}), \phi(t, \vec{x})] = 0$
- $[\phi(t, \vec{y})^2, \phi(t, \vec{x})] = \{\phi(t, \vec{y}), [\phi(t, \vec{y}), \phi(t, \vec{x})]\} = 0$

$$\therefore \dot{\phi}(t, \vec{x}) = -\frac{1}{i\hbar} [H[\phi, \pi; t], \phi(t, \vec{x})] = -\frac{1}{i\hbar} \int d^3 \vec{y} (-i\hbar c^2 \delta^3(\vec{y} - \vec{x}) \pi(t, \vec{y})) = c^2 \pi(t, \vec{x})$$

The commutator relation with $\pi(t, \vec{x})$ is a bit more tedious:

- $[\pi(t, \vec{y})^2, \pi(t, \vec{x})] = \{\pi(t, \vec{y}), [\pi(t, \vec{y}), \pi(t, \vec{x})]\} = 0$
- $[\phi(t, \vec{y})^2, \pi(t, \vec{x})] = \{\phi(t, \vec{y}), [\phi(t, \vec{y}), \pi(t, \vec{x})]\} = 2i\hbar\delta^3(\vec{y} - \vec{x})\phi(t, \vec{y})$

$$\begin{aligned}
& \bullet \left[\int d^3 \vec{y} \left(\vec{\nabla}_y \phi(t, \vec{y}) \right)^2, \pi(t, \vec{x}) \right] \\
&= \left[\oint_{bdry} d^2 \vec{S}_y \cdot \left(\phi(t, \vec{y}) \vec{\nabla}_y \phi(t, \vec{y}) \right) - \int d^3 \vec{y} \phi(t, \vec{y}) \vec{\nabla}_y^2 \phi(t, \vec{y}), \pi(t, \vec{x}) \right] \\
&= - \int d^3 \vec{y} \left(\phi(t, \vec{y}) \vec{\nabla}_y^2 [\phi(t, \vec{y}), \pi(t, \vec{x})] + [\phi(t, \vec{y}), \pi(t, \vec{x})] \cdot \vec{\nabla}_y^2 \phi(t, \vec{y}) \right) \\
&= -i\hbar \int d^3 \vec{y} \left(\phi(t, \vec{y}) \vec{\nabla}_y^2 \delta^3(\vec{y} - \vec{x}) + \delta^3(\vec{y} - \vec{x}) \vec{\nabla}_y^2 \phi(t, \vec{y}) \right) \\
&= -i\hbar \left(\oint_{bdry} d^2 \vec{S}_y \cdot \left(\phi(t, \vec{y}) \vec{\nabla}_y \delta^3(\vec{y} - \vec{x}) \right) - \int d^3 \vec{y} \vec{\nabla}_y \phi(t, \vec{y}) \cdot \vec{\nabla}_y \delta^3(\vec{y} - \vec{x}) + \vec{\nabla}^2 \phi(t, \vec{x}) \right) \\
&= i\hbar \left(\int d^3 \vec{y} \vec{\nabla}_y \phi(t, \vec{y}) \cdot \vec{\nabla}_y \delta^3(\vec{y} - \vec{x}) - \vec{\nabla}^2 \phi(t, \vec{x}) \right) \\
&= i\hbar \left(\oint_{bdry} d^2 \vec{S}_y \cdot \left(\delta^3(\vec{y} - \vec{x}) \vec{\nabla}_y \pi(t, \vec{y}) \right) - \int d^3 \vec{y} \delta^3(t, \vec{y}) \cdot \vec{\nabla}_y^2 \phi(t, \vec{y}) - \vec{\nabla}^2 \phi(t, \vec{x}) \right) \\
&= -2i\hbar \vec{\nabla}^2 \phi(t, \vec{x})
\end{aligned}$$

$$\begin{aligned}
\therefore \dot{\phi}(t, \vec{x}) &= -\frac{1}{i\hbar} [H[\phi, \pi; t], \pi(t, \vec{x})] \\
&= -\frac{1}{i\hbar} \left(\int d^3 \vec{y} \frac{1}{2} \left(\frac{mc}{\hbar} \right)^2 \cdot 2i\hbar \delta^3(\vec{y} - \vec{x}) \phi(t, \vec{y}) + \frac{1}{2} \left(-2i\hbar \vec{\nabla}^2 \phi(t, \vec{x}) \right) \right) \\
&= \left(\vec{\nabla}^2 - \left(\frac{mc}{\hbar} \right)^2 \right) \phi(t, \vec{x})
\end{aligned}$$

Combining these two, we recover the Klein-Gordon equation:

$$\left(\partial_\mu \partial^\mu - \left(\frac{mc}{\hbar} \right)^2 \right) \phi(t, \vec{x}) = 0.$$